ME2102: Engineering Innovation and Modelling

Gears

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Course Goals

- Survey of useful design components
- Compute forces and stresses relevant to component function
- Predict static and fatigue failure

Learning Outcomes

- Able to do calculations on design components (gears, screws, springs, bolts, welds, shafts, & bearings)

Calendar

Blue corresponds to	**Lectures**	given in LT7A
Red corresponds to	**CAD Tutorials**	given in the E2 Computer Clusters
Green corresponds to	**Design Tutorials**	given in E3-06-02/03/04
Yellow correspond to	**Assessments**	given in various locations

7	M	28 Sep	13–14	**L11:** Gears	
	W	30 Sep	12–14	**L12:** Power Screws	**DT1**
8	M	5 Oct	13–14	**L13:** Stress & Equilibrium	
	W	7 Oct	12–14	**L14:** Springs	**DT2***
9	M	12 Oct	13–14	**L15:** Bolts	
	W	14 Oct	12–14	**L16:** Welds	**DT3**
10	M	19 Oct	13–14	**L17:** Static Design Q&A	
	W	21 Oct	12–14	**Static Design CA Test** (Eng Aud) (30%)	
11	M	26 Oct	13–14	**L18:** Fatigue	
	W	28 Oct	12–14	**L19:** Shafts	**DT4**
12	M	2 Nov	13–14	**L20:** Bearings	
	W	4 Nov	12–14	**L21:** Fatigue Design Q&A	**DT5***
13	M	9 Nov	Deepavali		
	W	11 Nov	12–14	**Fatigue Design CA Test** (LTA) (20%)	

Tutorial Handouts

Gears

Concepts

- Gears are mechanical devices that transfer rotation from one power shaft to another via interlocking teeth, commonly used to exchange a decrease in angular speed ω for an increase in torque T. The **efficiency** of gears is given by $\eta = P_{out}/P_{in} = (T_{out}\omega_{out})/(T_{in}\omega_{in})$.
- In this class, we will discuss the design of spur gears and helical gears. **Helical gears** are characterized by two mating cylinders having parallel rotation axes, with teeth cut on the outside of the cylinders that mesh together to transfer torque and rotation. The angle that teeth are cut relative to the rotation axis is called the **helix angle** ψ. A **spur gear** is a helical gear with $\psi = 0$.
- Since the teeth (and thus the outer diameters) of two gears are designed to overlap, we define a gear's **pitch circle** upon which to base calculations. Two gears will mate correctly when their pitch circles are tangent.
- Along with ψ, a helical gear is chiefly characterized by a **pitch diameter** d (the diameter of its pitch circle), a transverse pressure angle ϕ (the angle at which two mating gear teeth make contact relative to the line connecting their centers), and an integer **number of teeth** N.
- For helical gears, sometimes a **normal pressure angle** ϕ_n is provided instead of ϕ. They are related by $\cos\psi = \frac{\tan\phi_n}{\tan\phi}$.
- Two mating gears must have equal **circular pitch** p (the distance along the pitch circle between corresponding points from two adjacent gear teeth) which is defined to be $\pi d/N$. The **diametral pitch** P must also be equal, and is defined as N/d.
- The smaller of two mating gears is typically called the **pinion**, while the larger is typically the **gear**. Given a pinion with N_P teeth and a gear with N_G teeth, their **gear ratio** is defined as $m = N_G/N_P \geq 1$. The angular speeds of a pinion and gear are related by $N_P\omega_P = N_G\omega_G$.
- **Interference** occurs when two gears make contact with corners of teeth rather than via normal surface contact, leading to wear and poor performance. To avoid interference, the number of teeth on the pinion must be at least:

$$N_P \geq \frac{2k\cos\psi\left(m + \sqrt{m^2 + (1+2m)\sin^2\phi}\right)}{(1+2m)\sin^2\phi}$$

  where $k = 1$ for **full-depth teeth** (default) and $k = 0.8$ for **stub teeth**; and the number of teeth on the gear can be at most:

$$N_G \leq \frac{N_P^2\sin^2\phi - 4k^2\cos^2\psi}{4k\cos\psi - 2N_P\sin^2\phi}$$

- A **rack** is a straight gear that can be thought of as a subset of a gear with infinite pitch diameter. A rack and pinion system effectively has gear ratio $m \to \infty$, so the pinion must satisfy:

$$N_P \geq \lim_{m\to\infty} \frac{2k\cos\psi\left(m + \sqrt{m^2 + (1+2m)\sin^2\phi}\right)}{(1+2m)\sin^2\phi} = \frac{2k\cos\psi}{\sin^2\phi}$$

Practice Problems

1) A 15-tooth spur pinion with 10 mm circular pitch rotates at 1600 rev/min to drive a 60-tooth gear. Determine the speed of the driven gear and their theoretical center-to-center distance.
2) Determine the smallest helical pinion with normal pressure angle 20° and helix angle 30° that will run with a rack without interference.
3) Determine the maximum gear ratio that can be achevied that avoids interference when combining a 25°-pressure angle 8-tooth stub pinion with a gear.
4) Determine the minimum pressure angle for a 9-tooth spur pinion to mesh with a rack while avoiding interference.

Equations Sheet

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CA II Equation Sheet

1 Power Screws

#	Equation
1.1	$d_r = d - p$
1.2	$d_m = d - p/2$
1.3	$\ell = np$
1.4	$w = p/2$
1.5	$t = p/2$
1.6	$\alpha_{sq} = 0°$
1.7	$\alpha_{ac} = 14.5°$
1.8	$T_R = \frac{Fd_m}{2}\left(\frac{\pi d_m f \sec\alpha + \ell}{\pi d_m - \ell f \sec\alpha}\right)$
1.9	$T_L = \frac{Fd_m}{2}\left(\frac{\pi d_m f \sec\alpha - \ell}{\pi d_m + \ell f \sec\alpha}\right)$
1.10	$T_C = F f_c d_c/2$
1.11	$e_R = \frac{F\ell}{2\pi(T_R + T_C)}$

2 Compression Springs

#	Equation
2.1	$F = kx$
2.2	$C = D/d$
2.3	$K_B = \frac{4C+2}{4C-3}$
2.4	$\tau_m = K_B \frac{8FD}{\pi d^3}$
2.5	$K_F = 0.45$
2.6	$\tau_y = K_F \frac{A}{d^m}$
2.7	$k = \frac{d^4 G}{8D^3 N_a}$
2.8	(1) $4 \le C \le 12$
2.9	(2) $3 \le N_a \le 15$
2.10	(3) $L_0 \le 5.26D$
2.11	(4) $x \le \frac{7}{8}(L_0 - L_s)$
2.12	(5) $n_s = 1.2 \le \frac{\tau_y}{\tau_m}$

Table 2.1: Spring end type parameters

Parameter	Plain	Ground	Squared	Squared & Ground
N_a	N_t	$N_t - 1$	$N_t - 2$	$N_t - 2$
p	$(L_0 - d)/N_a$	$L_0/(N_a + 1)$	$(L_0 - 3d)/N_a$	$(L_0 - d)/N_a$
L_s	$d(N_t + 1)$	dN_t	$d(N_t + 1)$	$d(N_t - 1)$

Table 2.2: Spring material properties

Material	A		m	Wire Diameter d		G	
	kpsi·in^m	MPa·mm^m	-	in	mm	Mpsi	GPa
Hard-Drawn wire (A227)	140	1783	0.190	$d < 0.032$	$d < 0.8$	11.7	80.7
				$0.032 \le d < 0.064$	$0.8 \le d < 1.6$	11.6	80.0
				$0.064 \le d < 0.125$	$1.6 \le d < 3.0$	11.5	79.3
				$0.125 \le d$	$3.0 \le d$	11.4	78.6
Music wire (A228)	201	2211	0.145	$d < 0.032$	$d < 0.8$	12.0	82.7
				$0.032 \le d < 0.064$	$0.8 \le d < 1.6$	11.85	81.7
				$0.064 \le d < 0.125$	$1.6 \le d < 3.0$	11.75	81.0
				$0.125 \le d$	$3.0 \le d$	11.6	80.0

3 Gears

#	Equation
3.1	$\eta = \frac{T_{out}\omega_{out}}{T_{in}\omega_{in}}$
3.2	$\psi_{spur} = 0°$
3.3	$\cos\psi = \frac{\tan\phi_n}{\tan\phi}$
3.4	$p = \frac{\pi d}{N}$
3.5	$P = \frac{N}{d}$
3.6	$m = \frac{N_G}{N_P}$
3.7	$N_P\omega_P = N_G\omega_G$
3.8	$k_{full} = 1$
3.9	$k_{stub} = 0.8$
3.10	$N_P \ge \frac{2k\cos\psi\left(m + \sqrt{m^2 + (1+2m)\sin^2\phi}\right)}{(1+2m)\sin^2\phi}$
3.11	$N_G \le \frac{N_P^2\sin^2\phi - 4k^2\cos^2\psi}{4k\cos\psi - 2N_P\sin^2\phi}$
3.12	Rack: $N_P \ge \frac{2k\cos\psi}{\sin^2\phi}$

4 Bolts

#	Equation
4.1	(i) $\sum F = 0$ (ii) $\sum M = 0$
4.2	Bearing: $\sigma = \frac{F}{A}$
4.3	Shear: $\tau = \frac{F}{A}$
4.4	Bending: $\sigma = \frac{My}{I}$
4.5	Torsion: $\tau = \frac{Tr}{J}$
4.6	(i) $A_\bigcirc = \frac{\pi d^2}{4}$ (ii) $A_\square = bh$
4.7	(i) $I_\bigcirc = \frac{\pi d^4}{64}$ (ii) $I_\square = \frac{bh^3}{12}$
4.8	(i) $J_\bigcirc = \frac{\pi d^4}{32}$ (ii) $J_\square = \frac{bh}{12}(b^2 + h^2)$
4.9	$\tau_y = \frac{\sigma_y}{\sqrt{3}}$
4.10	(i) $n_s = \frac{\tau_y}{\tau_m}$ (ii) $n_s = \frac{\sigma_y}{\sigma_m}$
4.11	(i) $F_i' = \frac{F}{N}$ (ii) $F_i'' = \frac{Mr_i}{\sum_i r_i^2}$

Table 4.1: Yield Stress for Standard Steel Bolts and Heat-Treated Steels

Bolt Class	σ_y [MPa]	Bolt Class	σ_y [kpsi]	Steel Class	σ_y [MPa]	σ_y [kpsi]
ISO 4.6	240	SAE Grade 1	36	1018 (HR)	220	32
ISO 5.8	420	SAE Grade 4	100	1018 (CD)	370	54
ISO 8.8	660	SAE Grade 7	115	1020 (HR)	210	30
ISO 9.8	720	SAE Grade 8	130	1020 (CD)	390	57

5 Welds

#	Equation	#	Equation	#	Equation	#	Equation
5.1	$\tau^* = \sqrt{\tau^2 + \left(\frac{\sigma_1 - \sigma_2}{2}\right)^2}$	5.2	$t = s/\sqrt{2}$	5.3	$\tau_y = 0.4\sigma_y$	5.4	$J = tJ_u, \quad I = tI_u$

5.5	|	(i) $J_u = \frac{h^3}{12}$	(ii) $I_u = \frac{h^3}{12}$	(iii) $y^* = \frac{h}{2}$
5.6	=	(i) $J_u = \frac{b(3h^2 + b^2)}{6}$	(ii) $I_u = \frac{bh^2}{2}$	(iii) $y^* = \frac{h}{2}$
5.7	⊓	(i) $J_u = \frac{8h^3 + 6hb^2 + b^3}{12} - \frac{h^4}{2h + b}$	(ii) $I_u = \frac{h^3(2b + h)}{3(b + 2h)}$	(iii) $y^* = \frac{h(b + h)}{b + 2h}$

Introduction

- What's a gear?

Rotating machine part typically used to transmit **rotational motion or torque** by means of a series of "teeth" that engage with compatible teeth of another gear or other part (Dictionary)



Toothed wheel that transmits power, torque, and speed between rotating shafts (Wikipedia)

Mechanical device that transfers rotation from one power shaft to another via interlocking teeth

Gears

- Types
- Terminology
- Pairs of gears
- Interference
- Maximum/minimum teeth
- Rack and pinion
- Solve problems for helical gear design

Gears









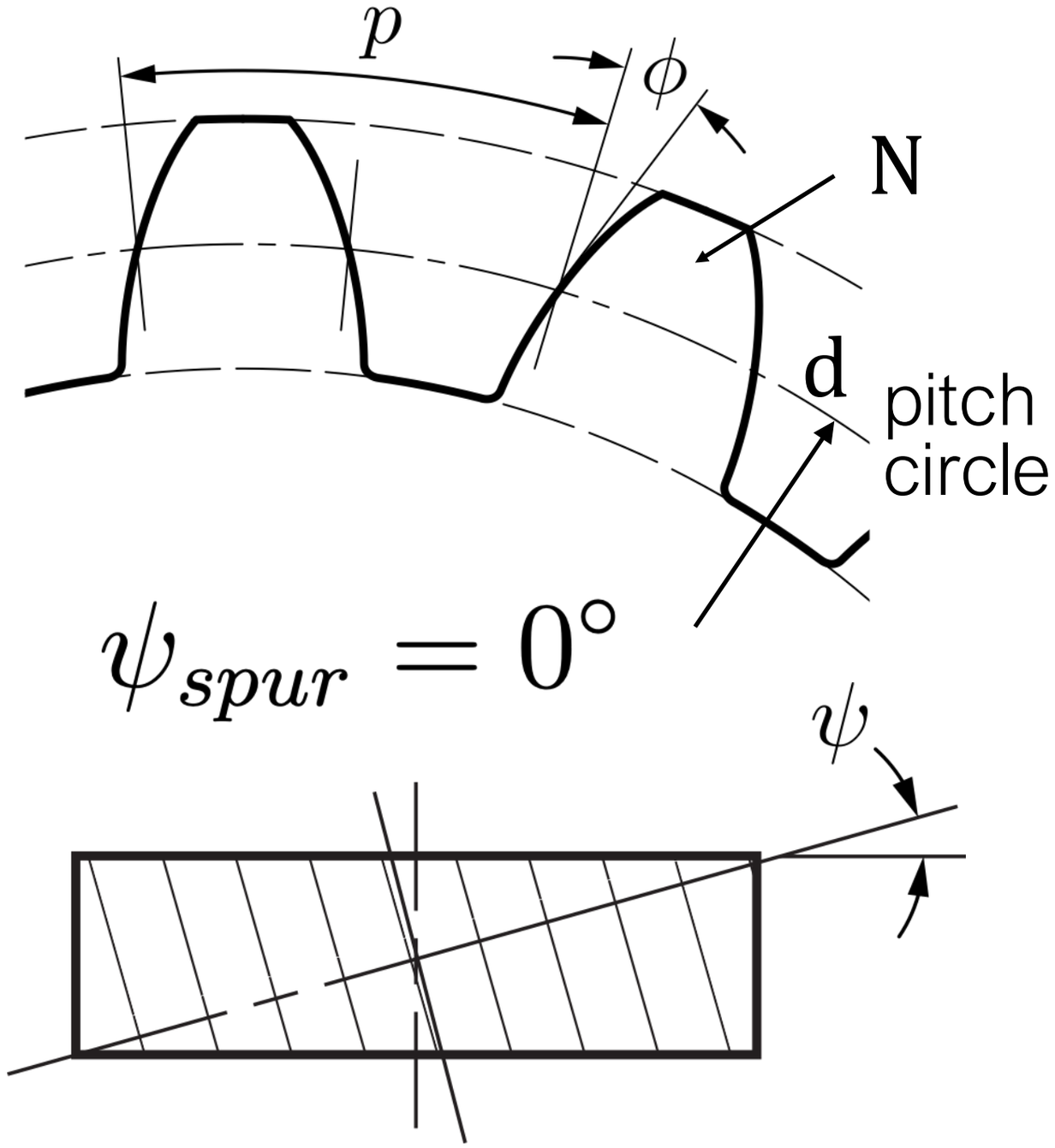






Terminology

- **d** pitch diameter
- **N** number of teeth
- **φ** pressure angle
- **p** circular pitch
- **P** diametral pitch
- **ψ** helix angle



$$p = \frac{\pi d}{N} \qquad P = \frac{N}{d}$$

$$\cos\psi = \frac{\tan\phi_n}{\tan\phi}$$

normal pressure angle

Pairs of Gears

- pitch must match
- smaller is **pinion**
- larger is **gear**
- **m** is gear ratio
- usually pinion drives gear

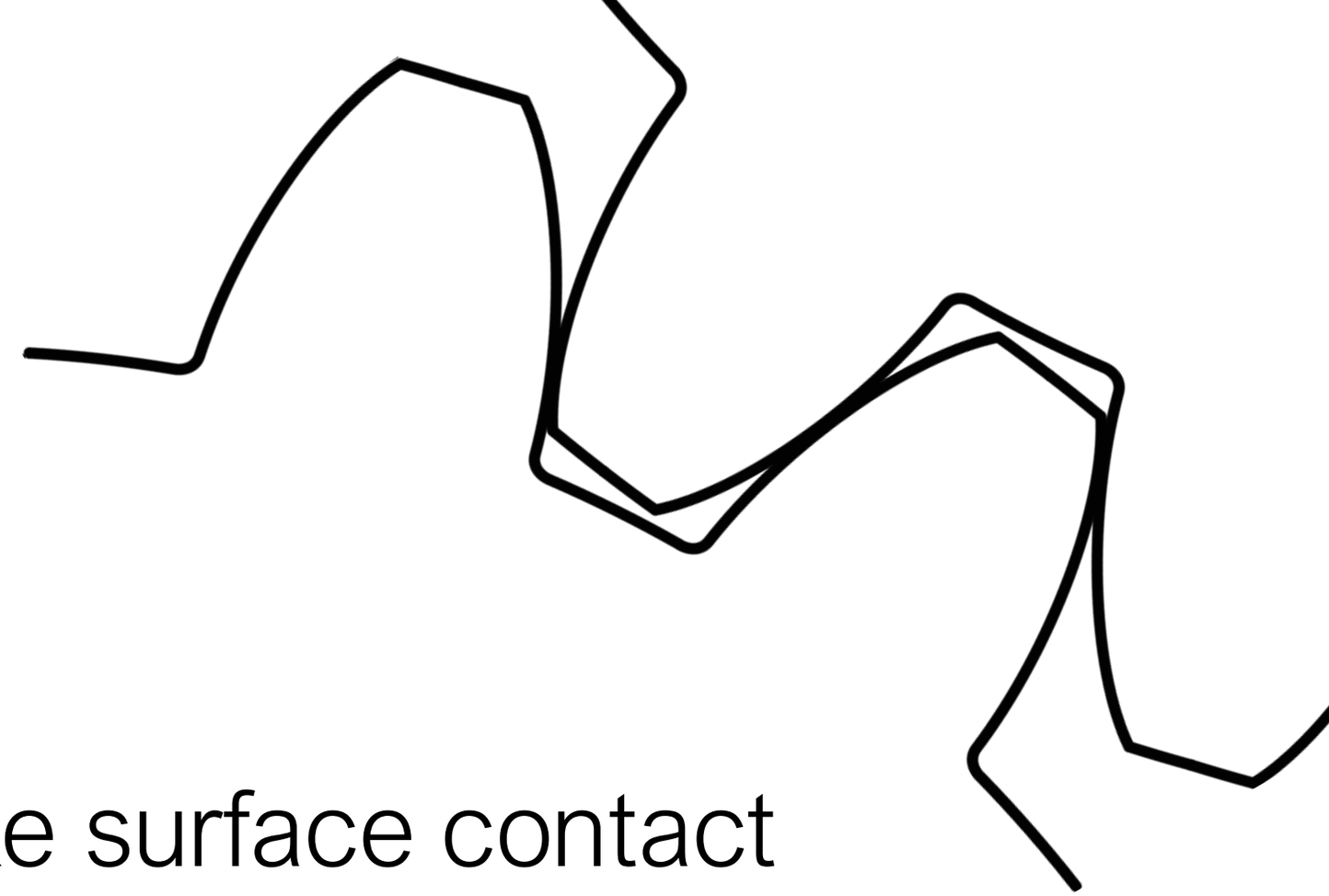
$$N_P \omega_P = N_G \omega_G$$

$$m = \frac{N_G}{N_P}$$

$$P = T\omega$$

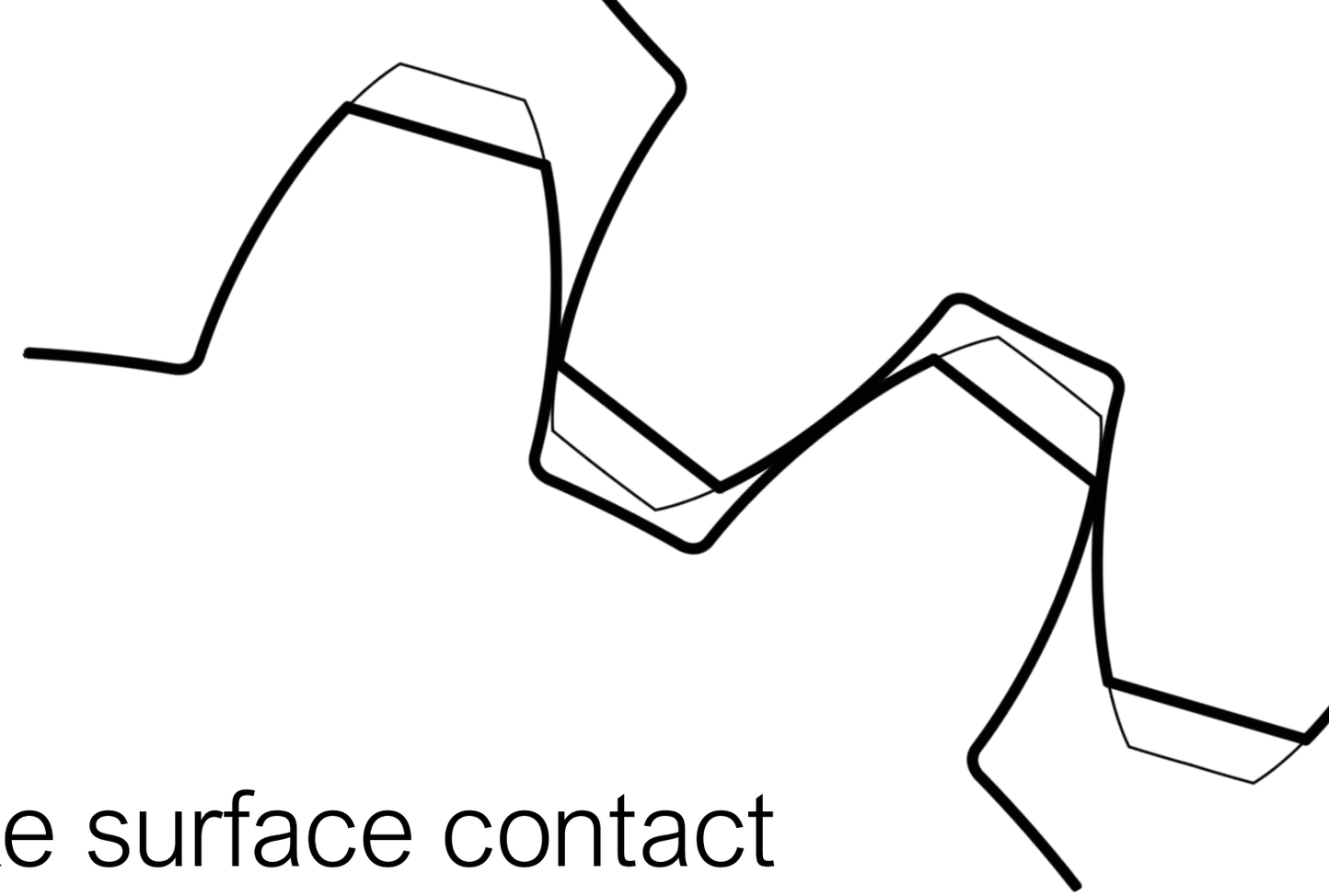
$$\eta = \frac{T_{out}\omega_{out}}{T_{in}\omega_{in}}$$

Interference



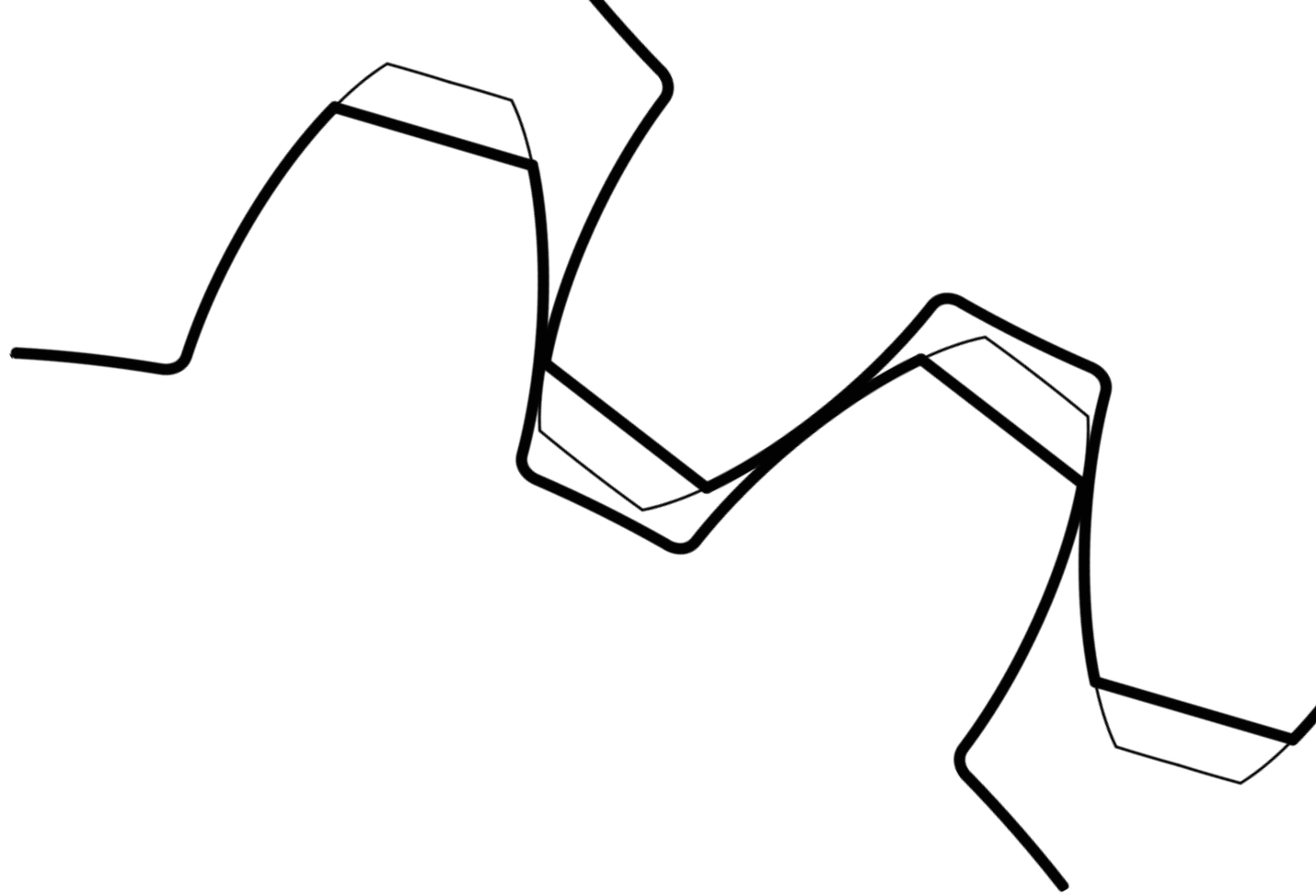
- Teeth should make surface contact
- **Interference** when contact at corners
- Leads to wear and damage
- # teeth, pressure angle, helix angle

Interference



- Teeth should make surface contact
- **Interference** when contact at corners
- Leads to wear and damage
- # teeth, pressure angle, helix angle
- **full-depth** teeth vs **stub** teeth

Interference



$$N_P \geq \frac{2k\cos\psi\left(m+\sqrt{m^2+(1+2m)\sin^2\phi}\right)}{(1+2m)\sin^2\phi}$$

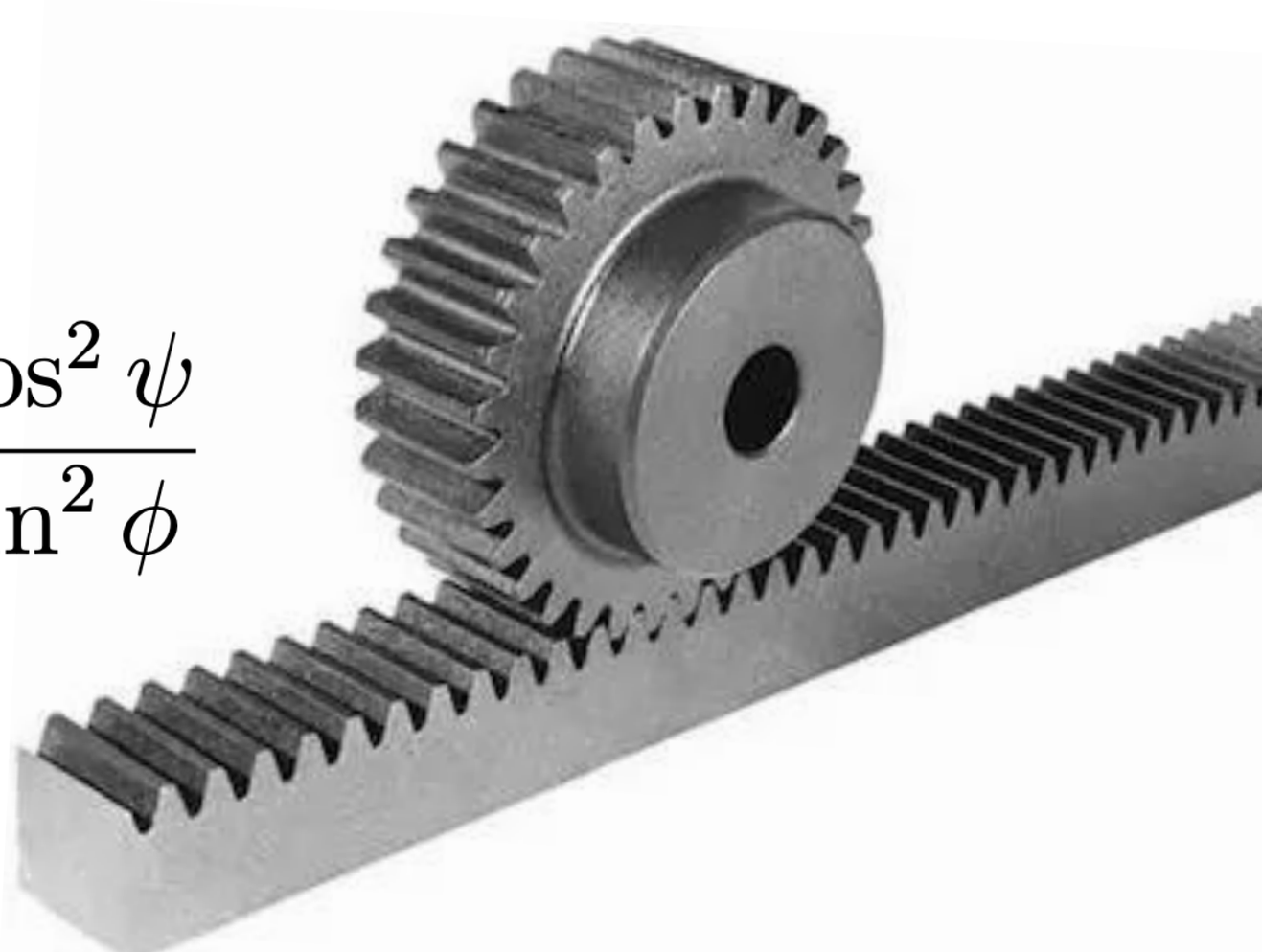
$$N_G \leq \frac{N_P^2\sin^2\phi - 4k^2\cos^2\psi}{4k\cos\psi - 2N_P\sin^2\phi}$$

$$k_{full} = 1$$

$$k_{stub} = 0.8$$

Rack and Pinion

$$N_G \leq \frac{N_P^2 \sin^2\phi - 4k^2\cos^2\psi}{4k\cos\psi - 2N_P\sin^2\phi}$$



$$N_P \geq \frac{2k\cos\psi\left(m + \sqrt{m^2 + (1+2m)\sin^2\phi}\right)}{(1+2m)\sin^2\phi}$$

$$\lim_{m\to\infty} = \frac{2k\cos\psi}{\sin^2\phi}$$

Example 1

A pinion rotating at 1120 rev/min drives a gear at a speed of 544 rev/min. The pinion has 17 teeth and a diametral pitch of 8 teeth/in. How far apart should the center of the pinion be placed relative to the center of the gear?

$$P = \frac{N_P}{d_P} = \frac{N_G}{d_G} \qquad d_P = \frac{N_P}{P} = \frac{17}{8 \text{ teeth/in}} = 2.125”$$

$$N_P \omega_P = N_G \omega_G \qquad N_G = N_P \frac{\omega_P}{\omega_G} = (17) \frac{1120 \text{ rev/min}}{544 \text{ rev/min}} = 35$$

$$d_G = \frac{N_G}{P} = \frac{35}{8 \text{ teeth/in}} = 4.375”$$

$$\frac{d_P + d_G}{2} = \frac{(2.125”) + (4.375”)}{2} = \boxed{3.25”}$$

Example 2

Find the smallest helical gear with $20°$ normal pressure angle and $30°$ helix angle that will run with itself while avoiding interference.

$$N_P \geq \frac{2k\cos\psi\left(m+\sqrt{m^2+(1+2m)\sin^2\phi}\right)}{(1+2m)\sin^2\phi} \qquad \begin{matrix} k=1 \\ m=1 \\ \psi=30° \end{matrix}$$

$$\phi_n = 20° \quad \cos\psi = \frac{\tan\phi_n}{\tan\phi} \quad \phi = \tan^{-1}\left(\frac{\tan 20°}{\cos 30°}\right) = 22.80°$$

$$\frac{2(1)\cos 30°\left(1+\sqrt{1^2+(1+2)\sin^2 22.80°}\right)}{(1+2)\sin^2 22.80°} = 8.48 \leq \boxed{9}$$

Example 3

Find the smallest helical gear with 20° normal pressure angle and 30° helix angle that will run with itself while avoiding interference.
What is the largest gear that will run with this pinion?

$$N_G \leq \frac{N_P^2 \sin^2\phi - 4k^2\cos^2\psi}{4k\cos\psi - 2N_P\sin^2\phi}$$

$$k = 1$$

$$\psi = 30^\circ$$

$$\phi = 22.80^\circ$$

$$N_P = 9$$

$$\frac{(9)^2\sin^2 22.80^\circ - 4(1)^2\cos^2 30^\circ}{4(1)\cos 30^\circ - 2(9)\sin^2 22.80^\circ} = 12.02 \geq \boxed{12}$$

Practice

$$N_P \geq \frac{2k\cos\psi}{\sin^2\phi}$$

What is the smallest spur pinion that will run with a rack with 30° pressure angle?



A) 4

B) 6

C) 8

D) 10

Practice

$$N_P \geq \frac{2k\cos\psi\left(m+\sqrt{m^2+(1+2m)\sin^2\phi}\right)}{(1+2m)\sin^2\phi}$$

For the same gear ratio and pressure angle, which specification will allow a pinion with the fewest teeth?



A) spur with full teeth

B) helical with full teeth

C) spur with stub teeth

D) helical with stub teeth

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